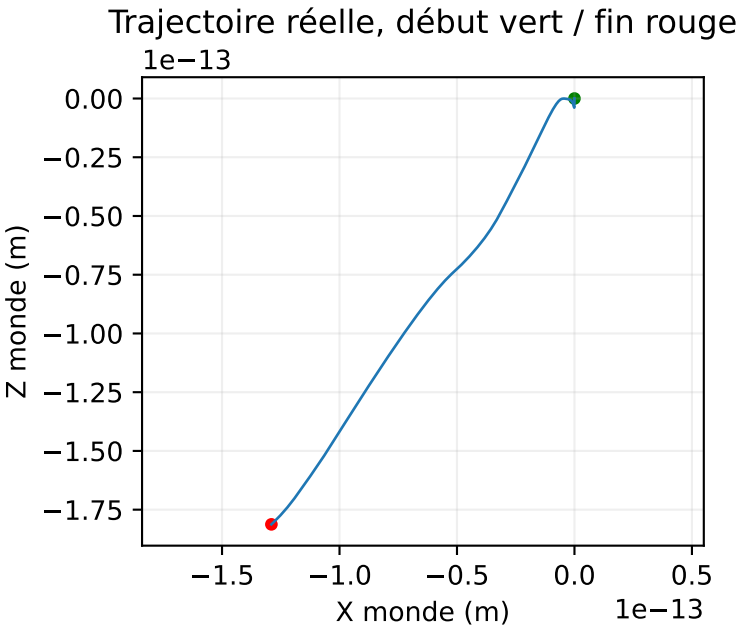
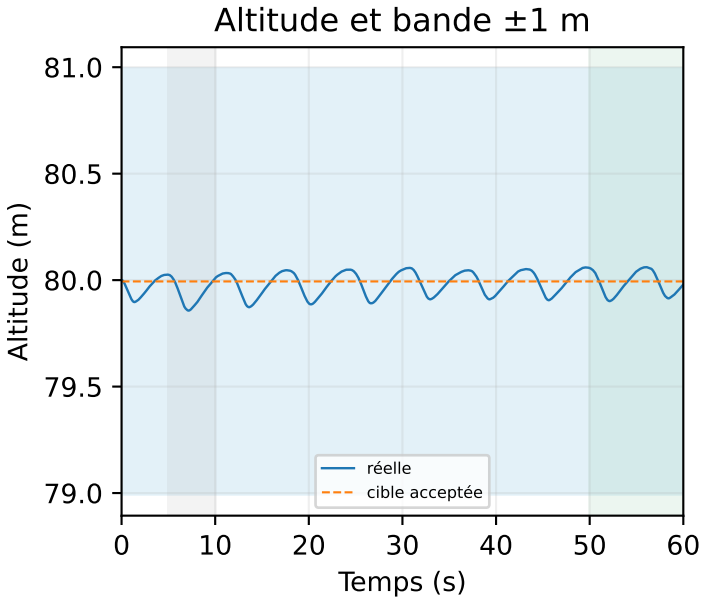
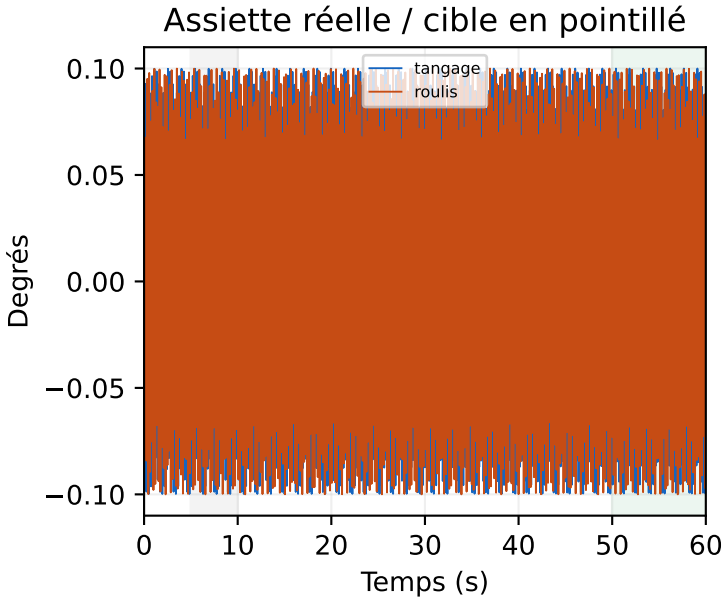
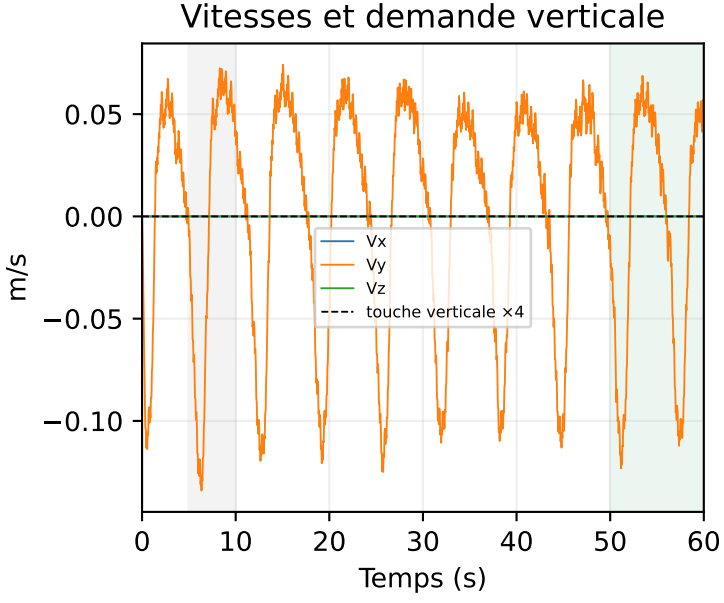
slow-radio / limited / hover — PASS

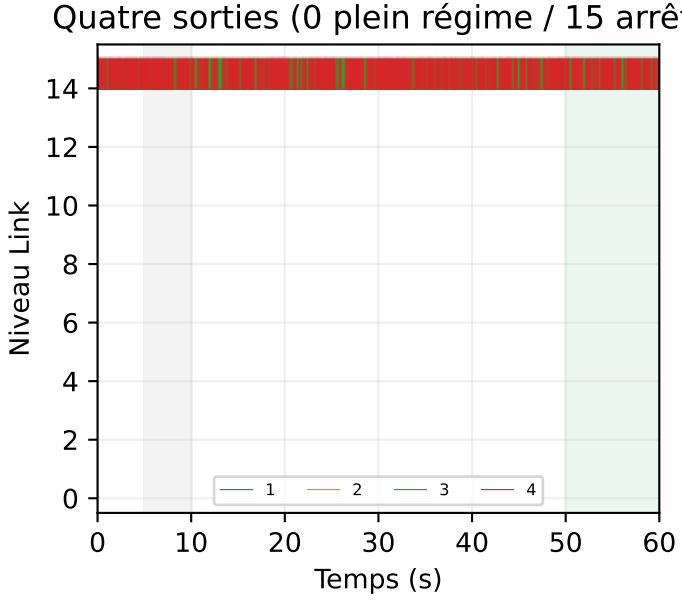
Maintien sans touche pendant toute la simulation.

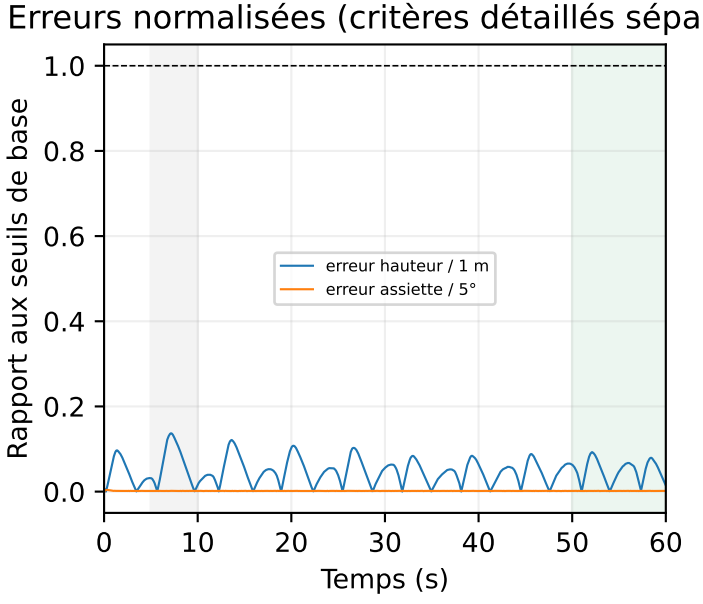










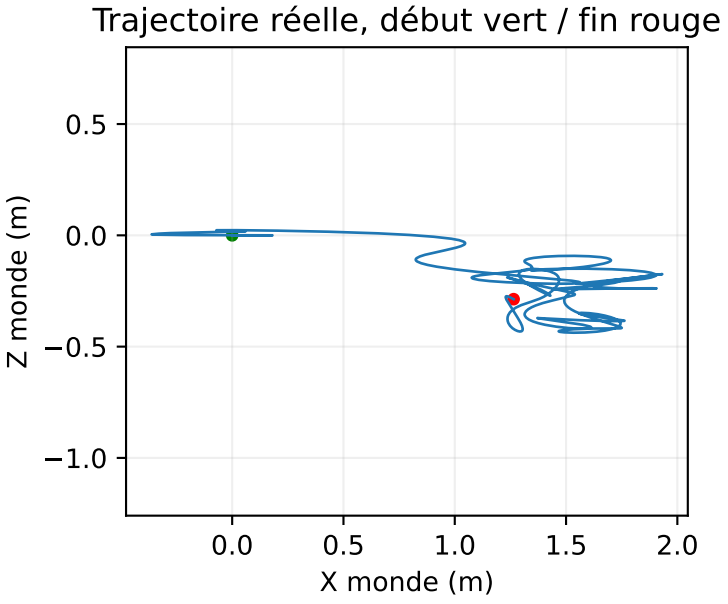


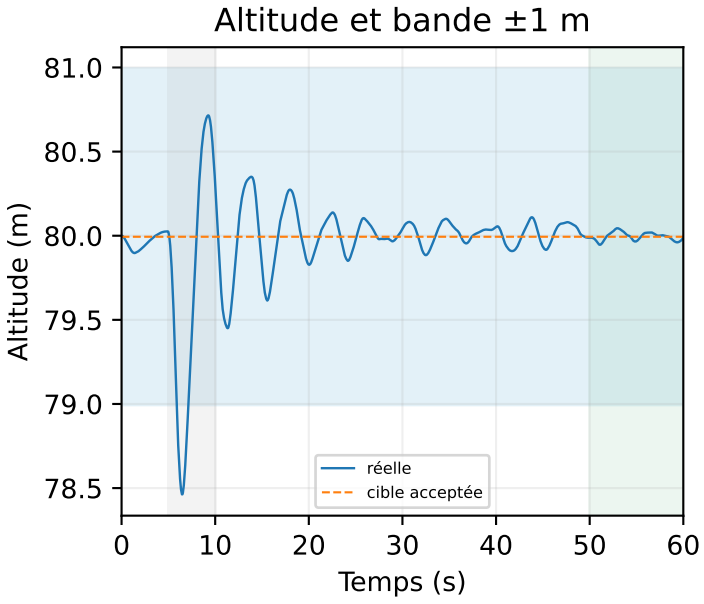
Erreur finale hauteur 0.014 m ; excursion finale 0.160 m ; assiette finale max 0.000° ; freinage — s. Critères échoués : aucun.

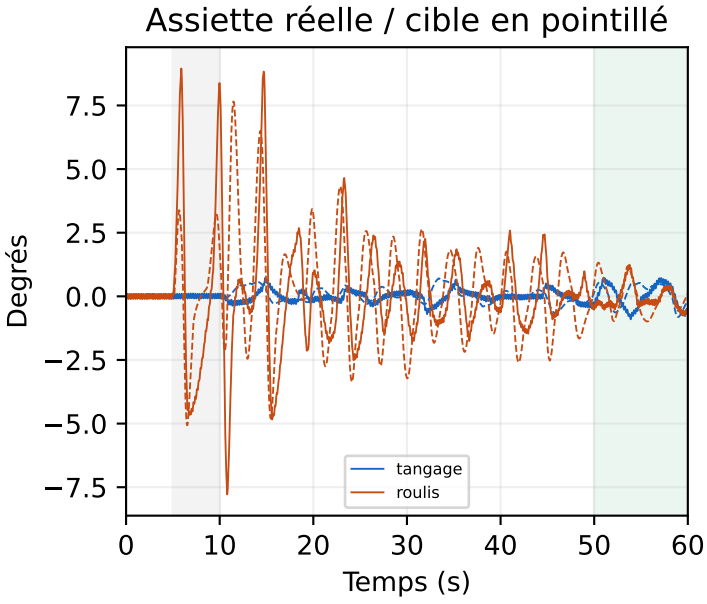
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

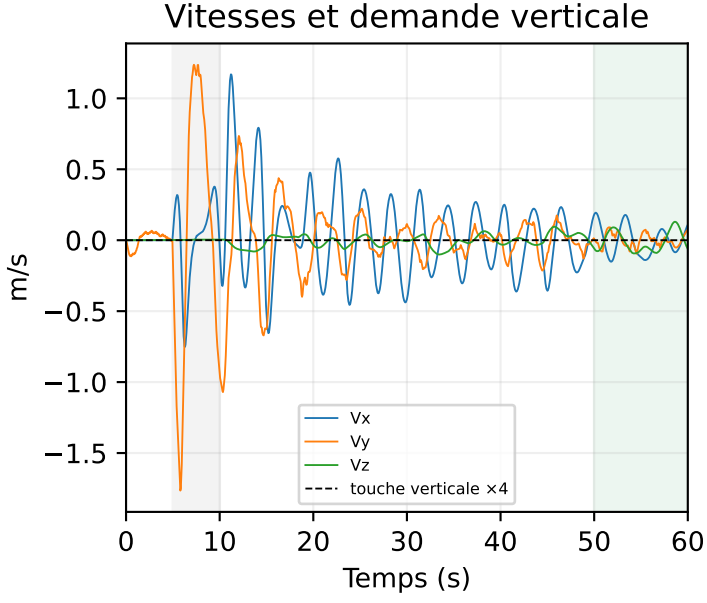
slow-radio / limited / load-swing — PASS

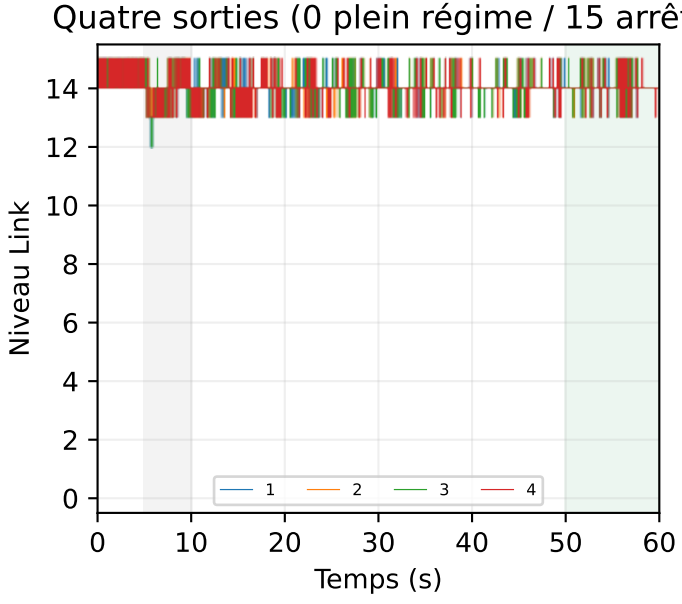
Aucune touche ; charge ponctuelle de 12 kg sur corde 5 m, inclinée initialement de 20°, ajoutée à 5 s.

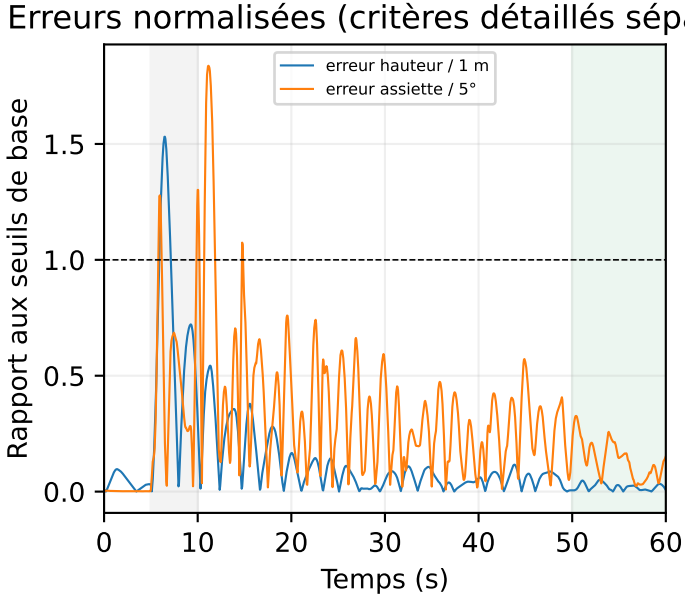










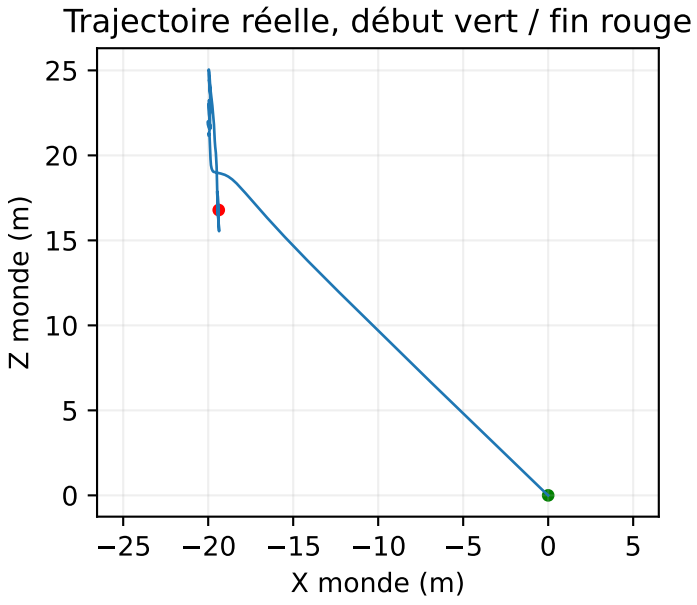


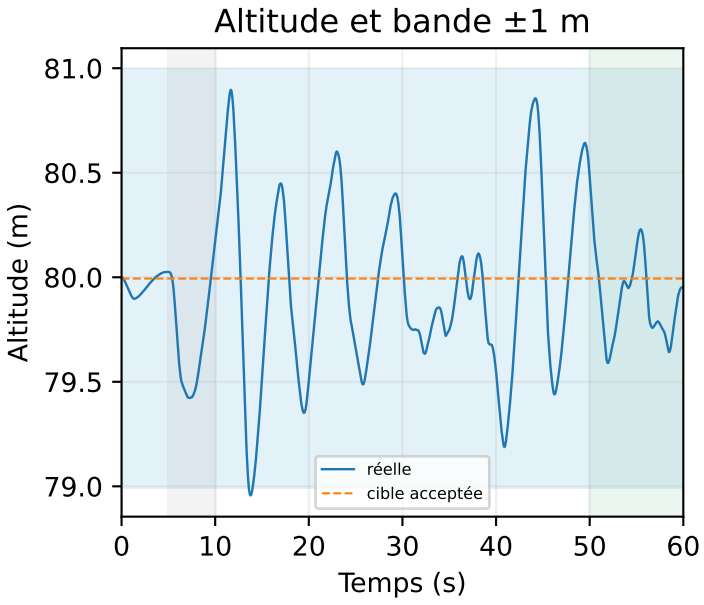
Erreur finale hauteur 0.007 m ; excursion finale 0.099 m ; assiette finale max 1.427° ; freinage — s. Critères échoués : aucun.

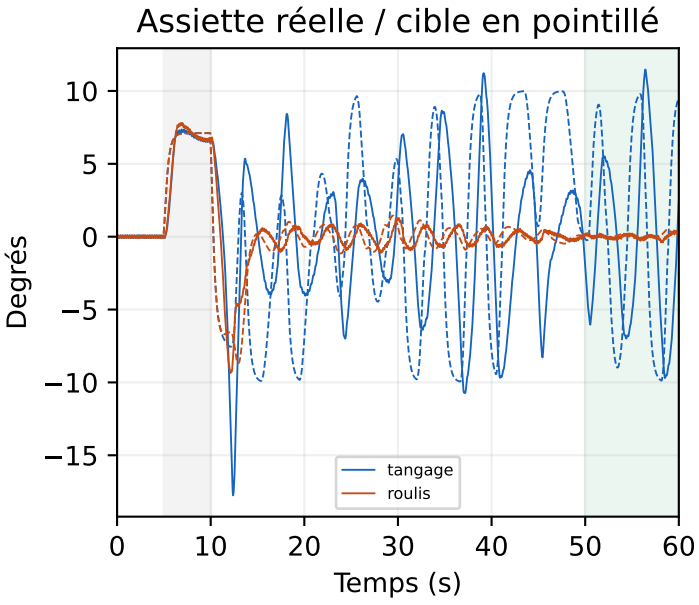
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

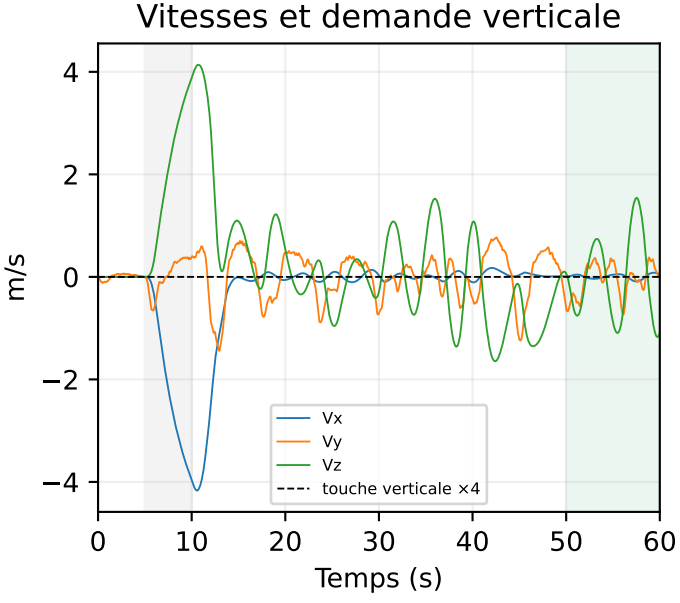
slow-radio / limited / forward-left — FAIL

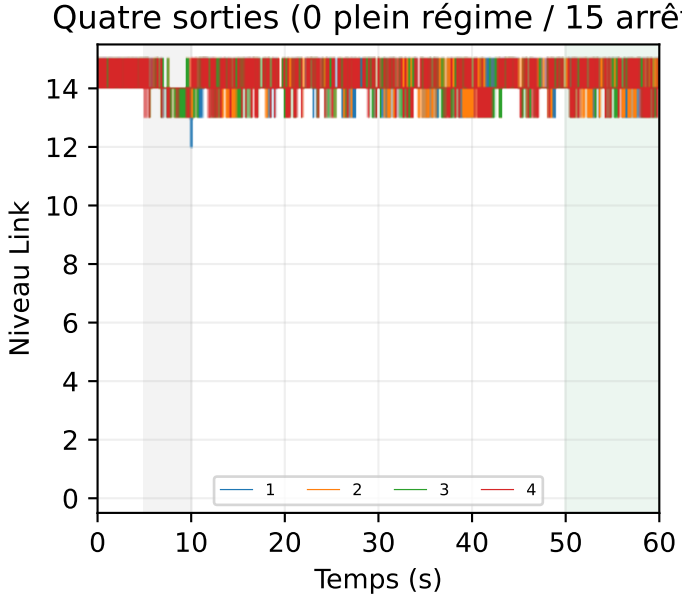
W+Q entre 5 et 10 s, mode déplacement latéral.

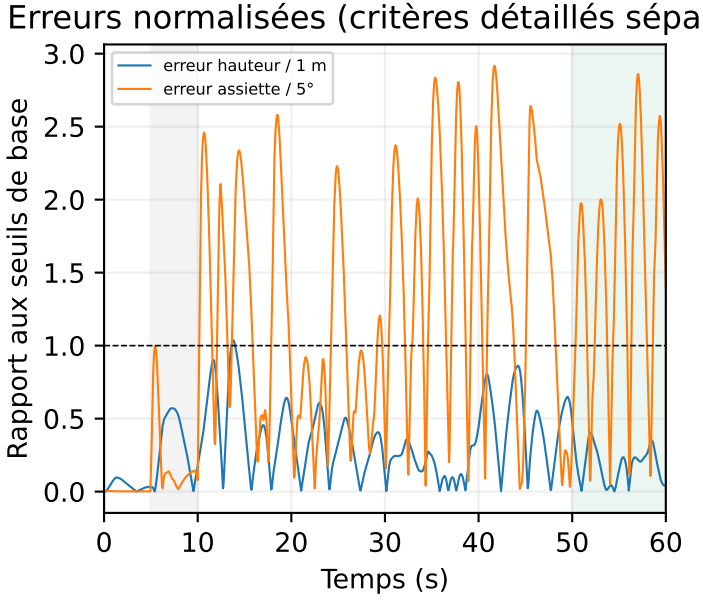










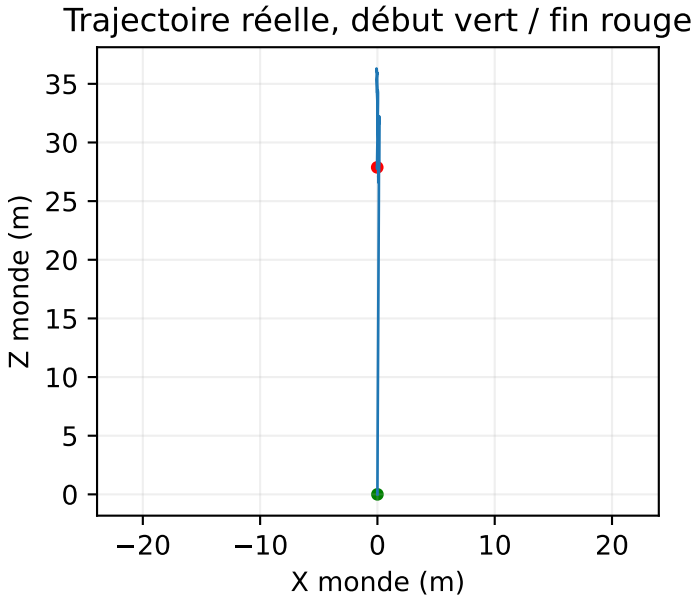


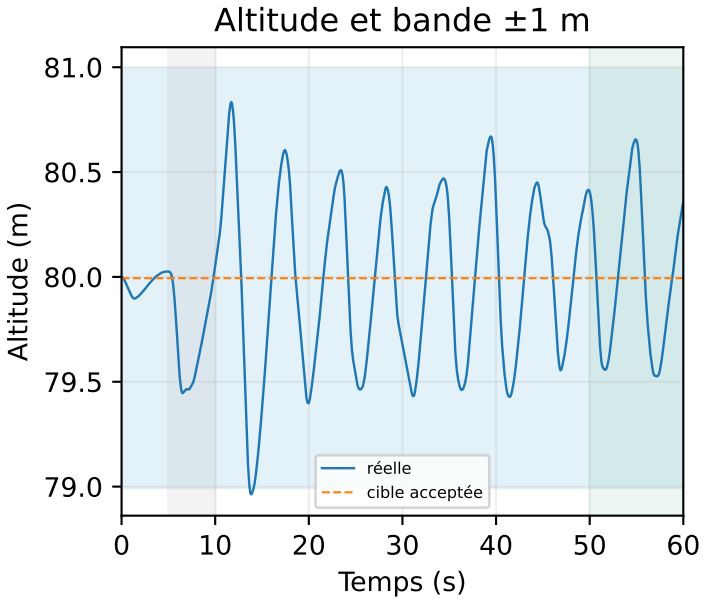
Erreur finale hauteur 0.042 m ; excursion finale 0.858 m ; assiette finale max 11.407° ; freinage 7.050 s. Critères échoués : tail_tilt, tail_speed_rms, final_speed, braking.

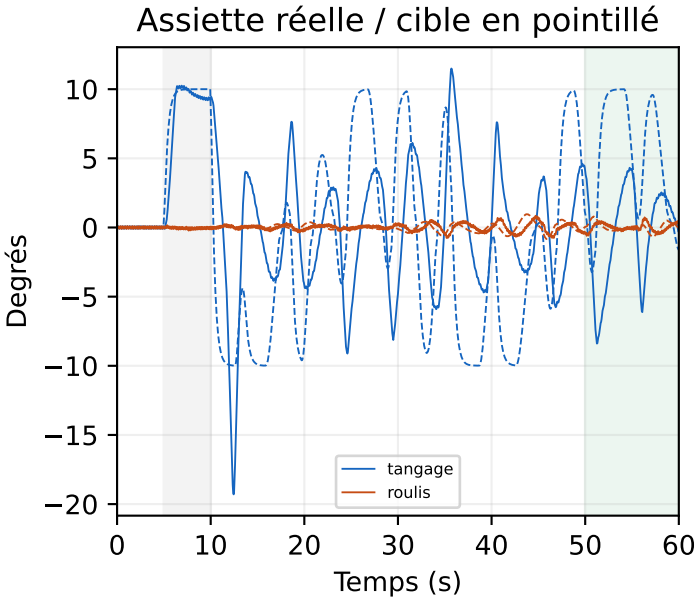
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

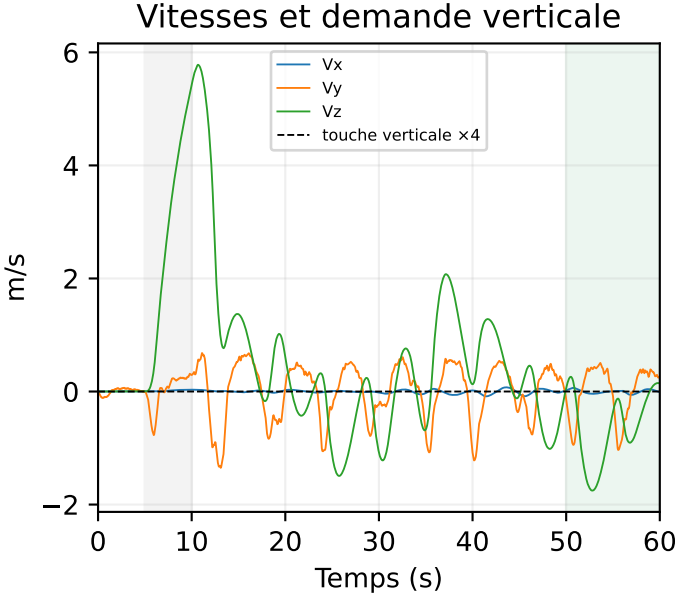
slow-radio / limited / brake — FAIL

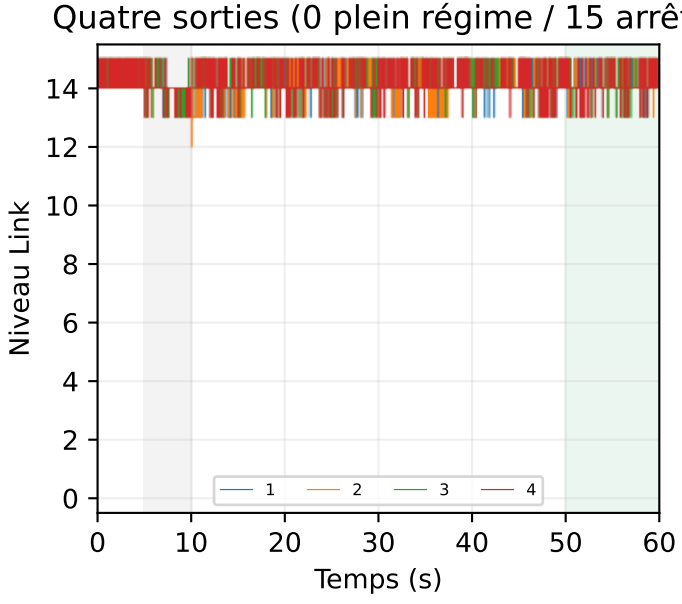
W de 5 à 10 s, relâchement, délai de freinage mesuré sur 15 observations consécutives.

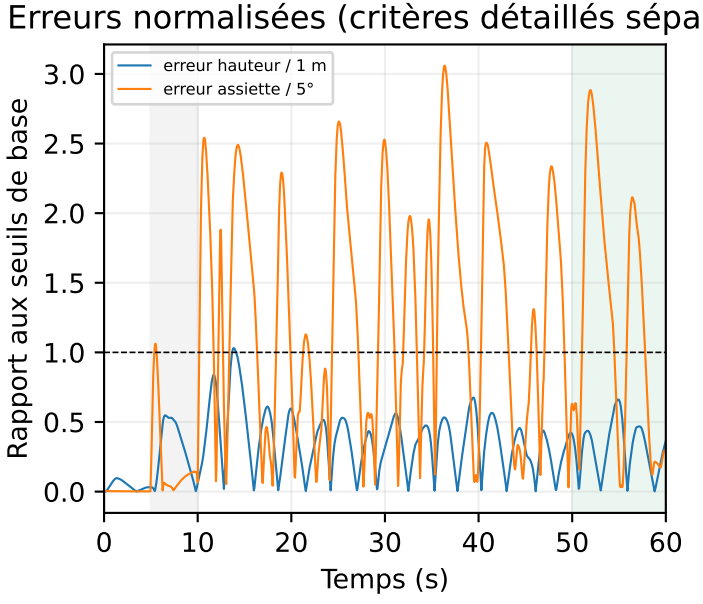










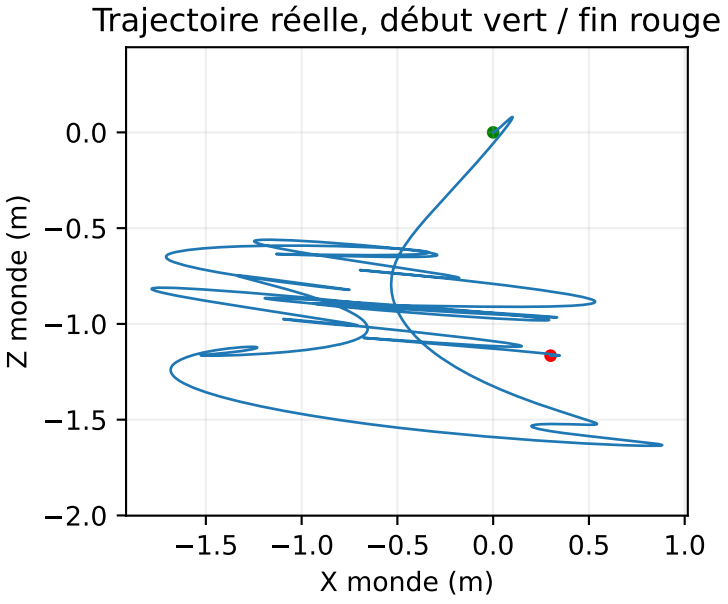


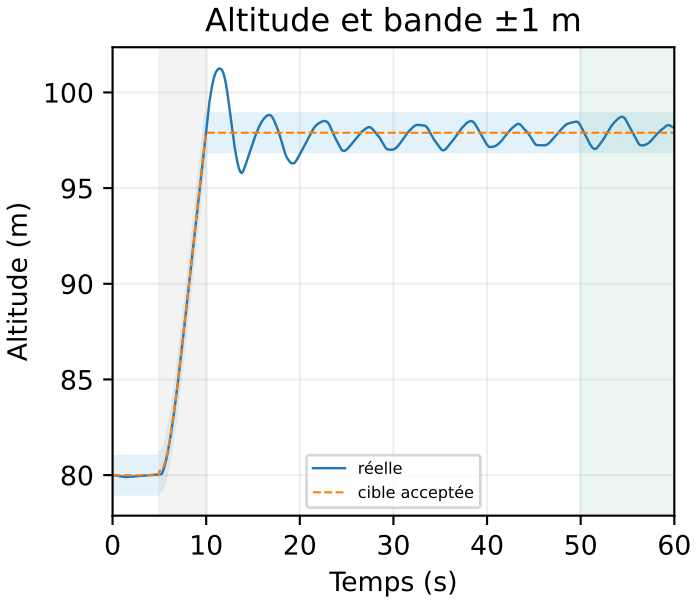
Erreur finale hauteur 0.365 m ; excursion finale 1.130 m ; assiette finale max 8.356° ; freinage 7.600 s. Critères échoués : tail_height_span, tail_tilt, tail_speed_rms, braking.

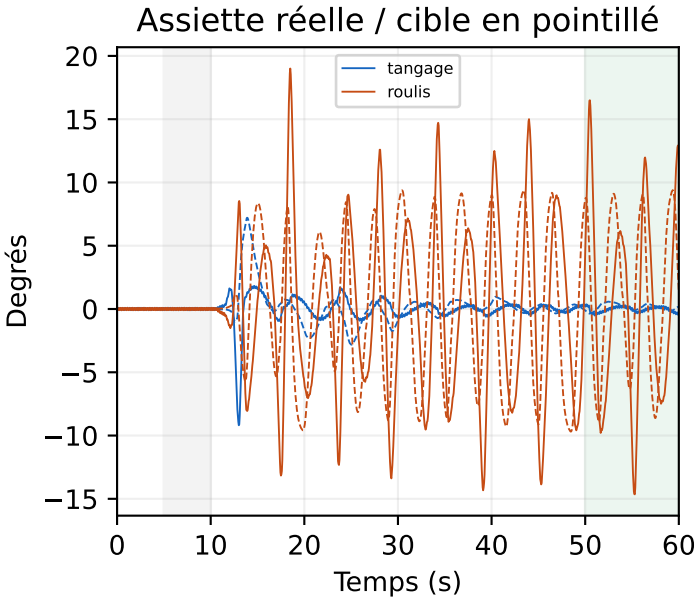
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

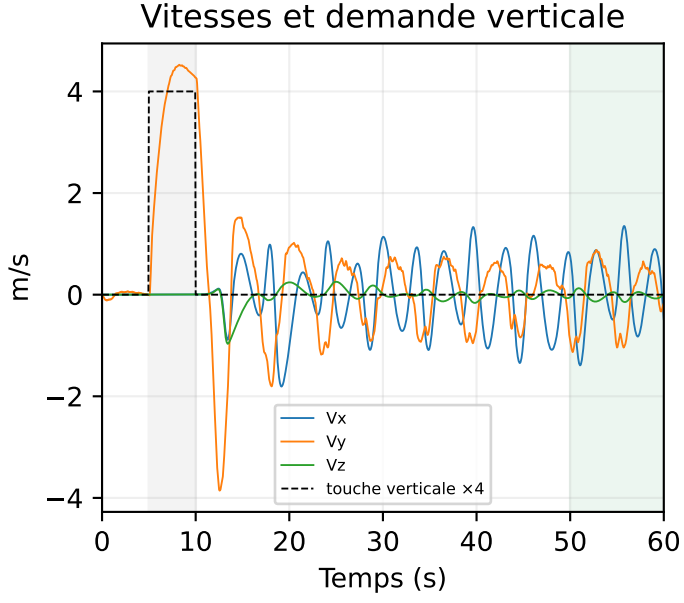
slow-radio / limited / climb — FAIL

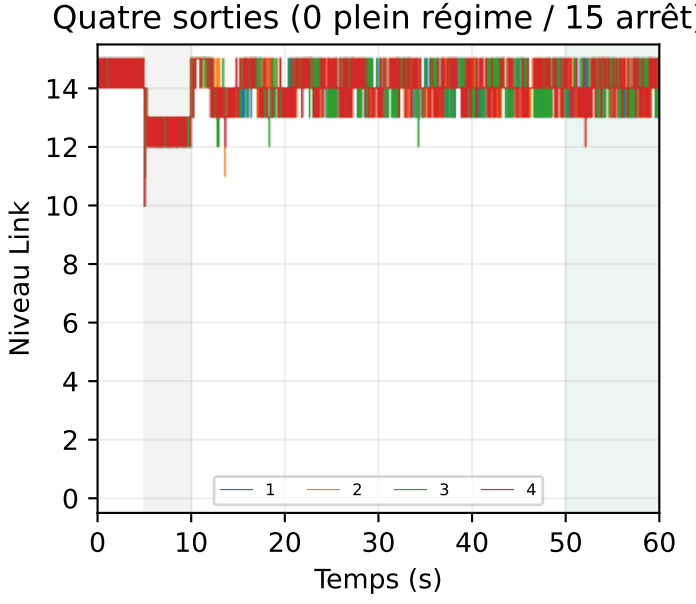
Espace entre 5 et 10 s, puis relâchement et maintien de la hauteur atteinte.

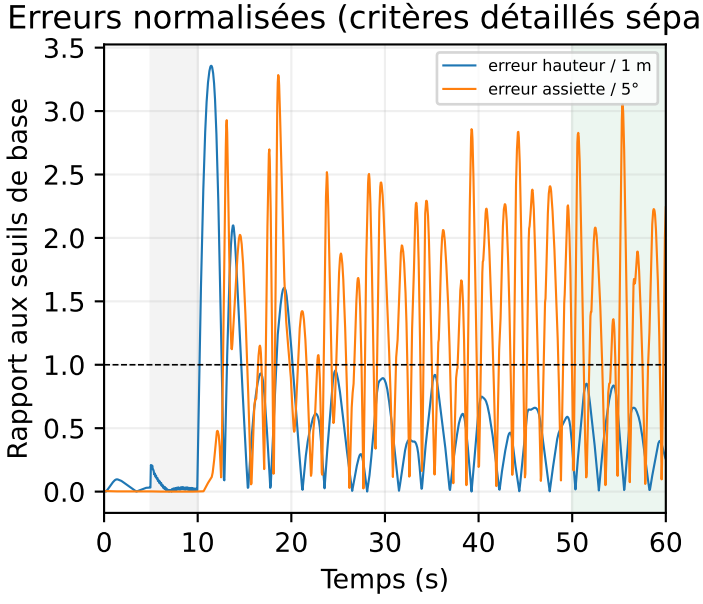










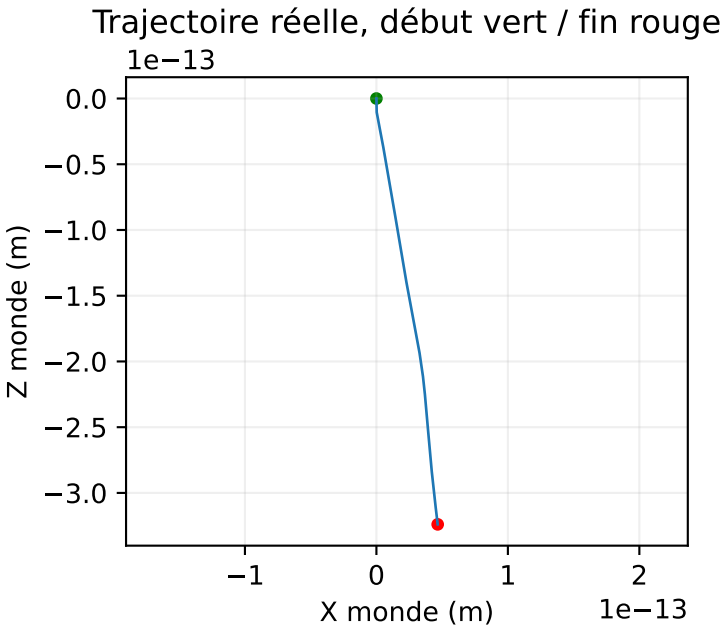


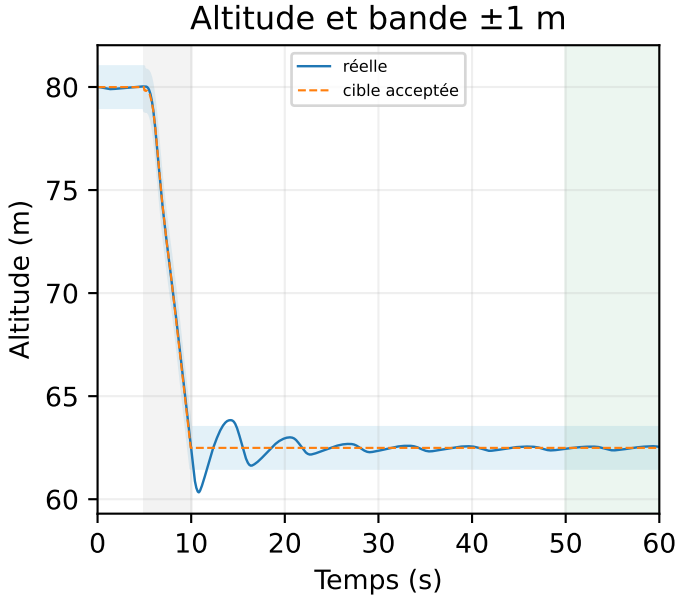
Erreur finale hauteur 0.256 m ; excursion finale 1.687 m ; assiette finale max 16.543° ; freinage — s. Critères échoués : tail_height_span, tail_tilt, tail_speed_rms, final_speed, transient_height.

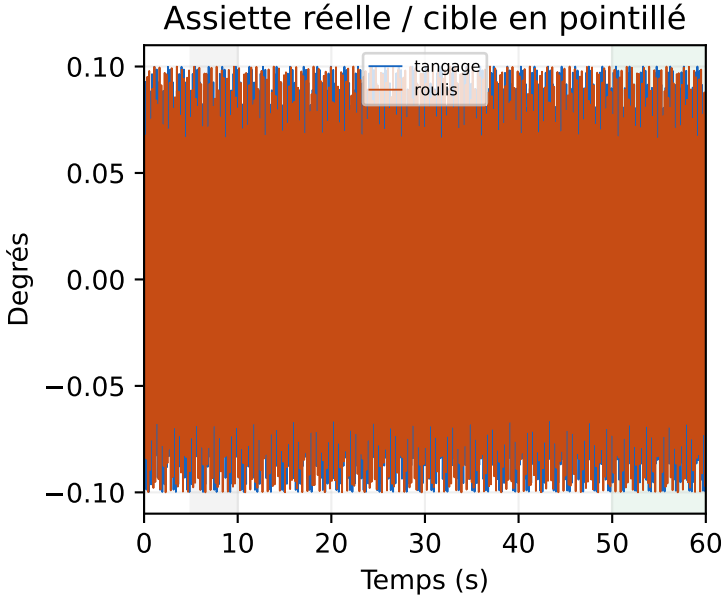
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

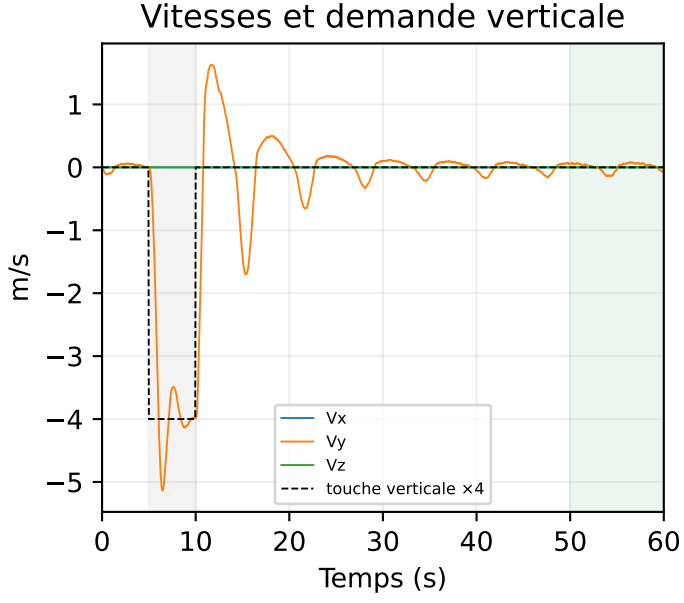
slow-radio / limited / descend — FAIL

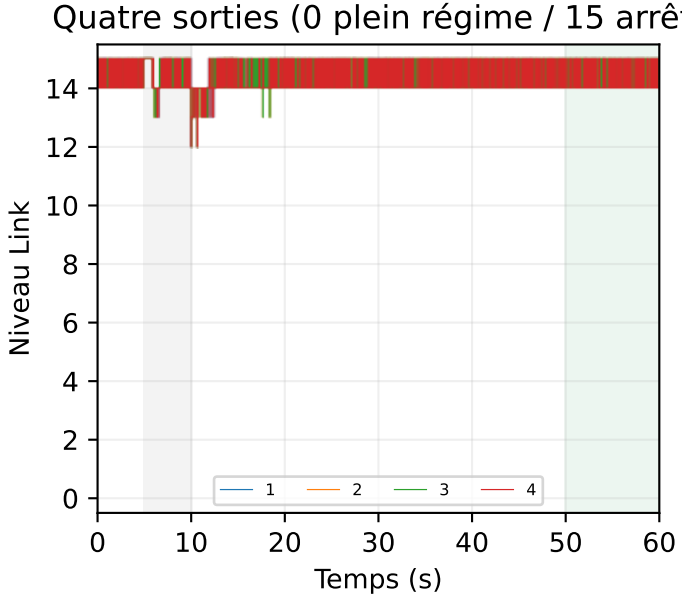
Shift entre 5 et 10 s, puis relâchement et maintien.

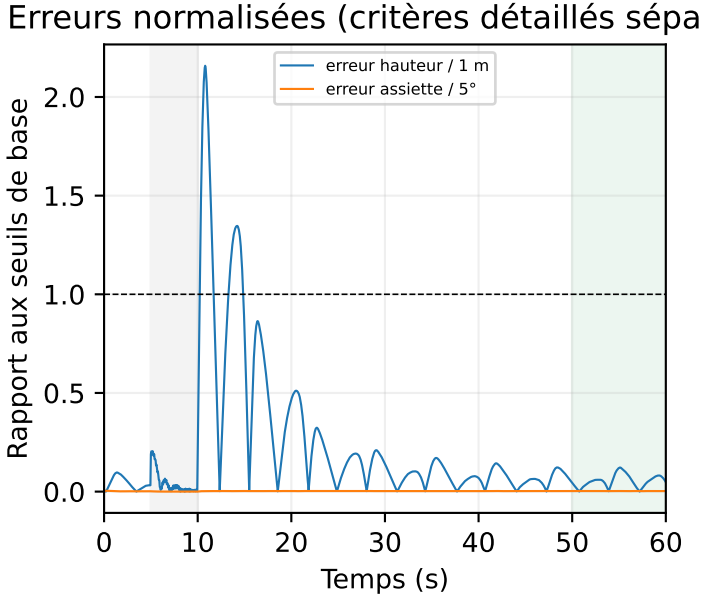












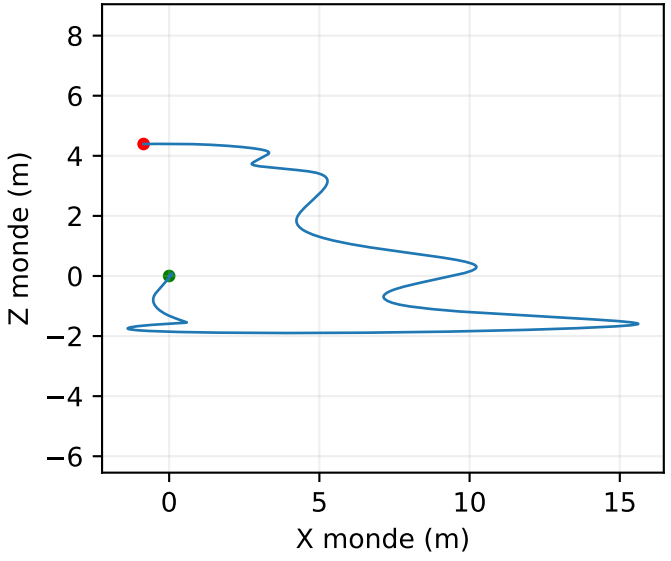
Erreur finale hauteur 0.048 m ; excursion finale 0.204 m ; assiette finale max 0.000° ; freinage — s. Critères échoués : transient_height.

Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

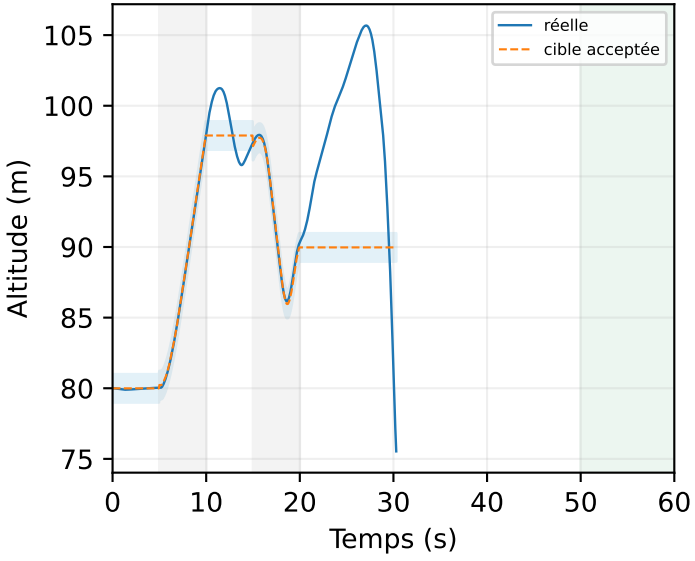
slow-radio / limited / vertical-cycle — FAIL

Espace de 5 à 10 s, maintien 10–15 s, Shift 15–20 s, puis maintien.

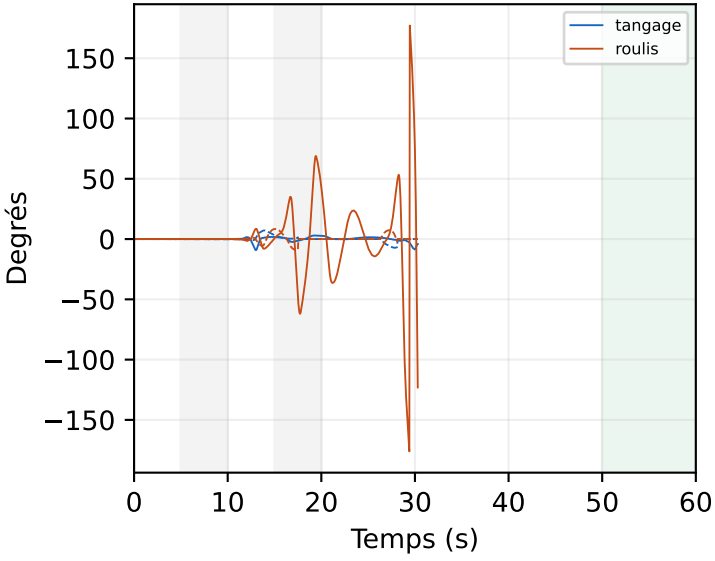
Trajectoire réelle, début vert / fin rouge



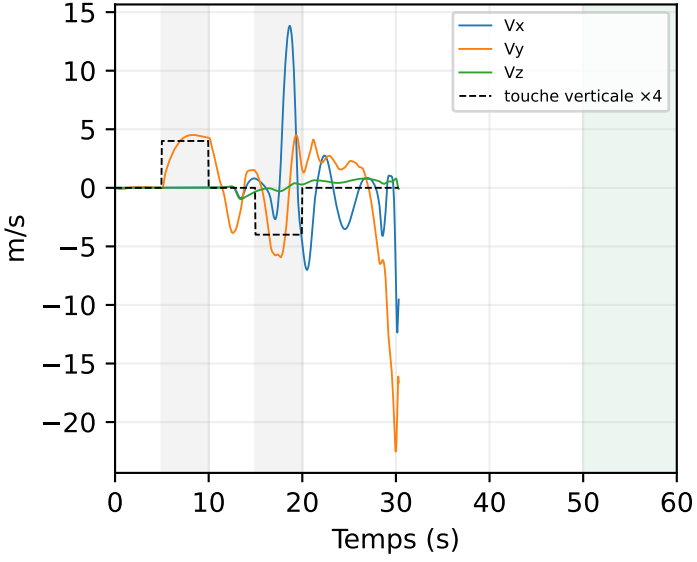
Altitude et bande ±1 m



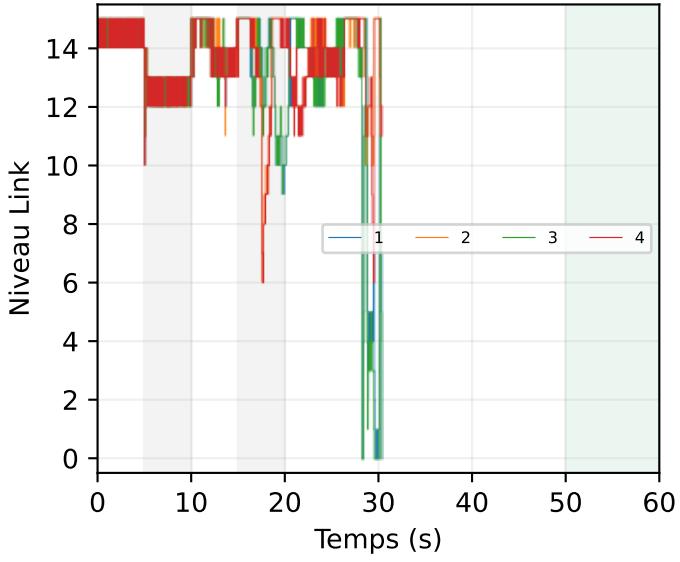
Assiette réelle / cible en pointillé



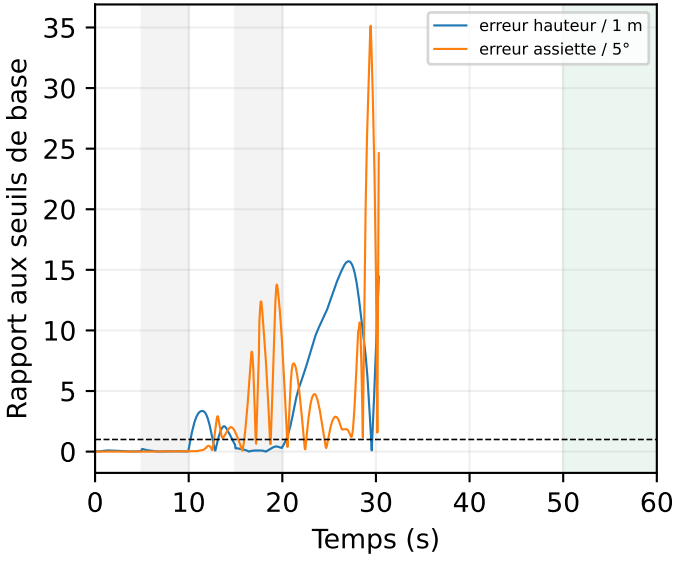
Vitesses et demande verticale



Quatre sorties (0 plein régime / 15 arrêt)



Erreurs normalisées (critères détaillés sépare

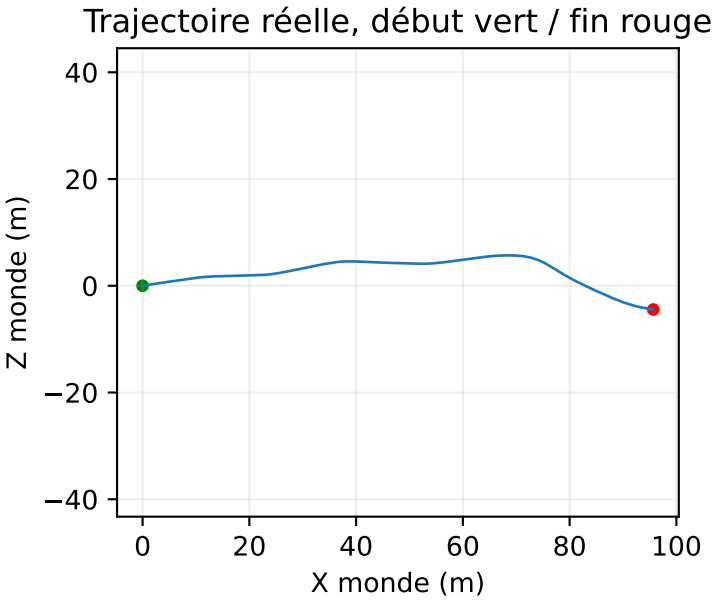


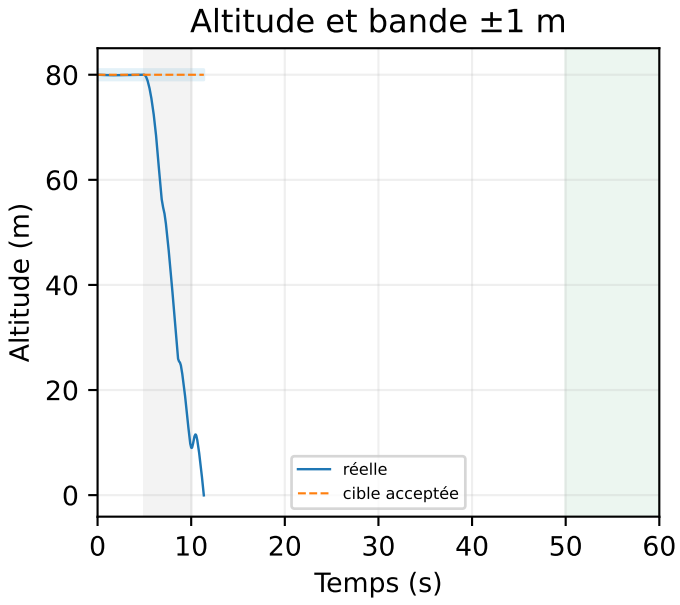
Erreur finale hauteur 14.439 m ; excursion finale — m ; assiette finale max —° ; freinage — s. Critères échoués : complete, tail_height_peak, tail_height_rms, tail_height_span, tail_tilt, tail_speed_rms, final_speed, transient_height, transient_tilt, active_attitude, active_vertical_rate. Arrêt : Poussee indisponible

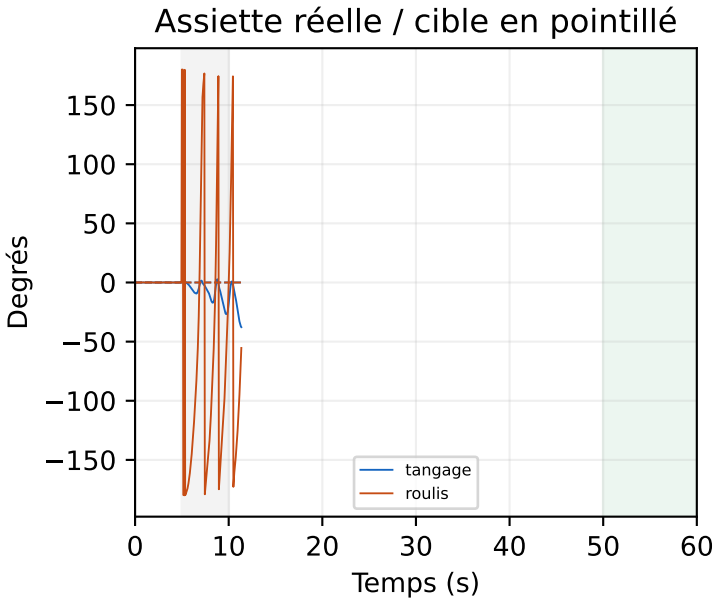
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

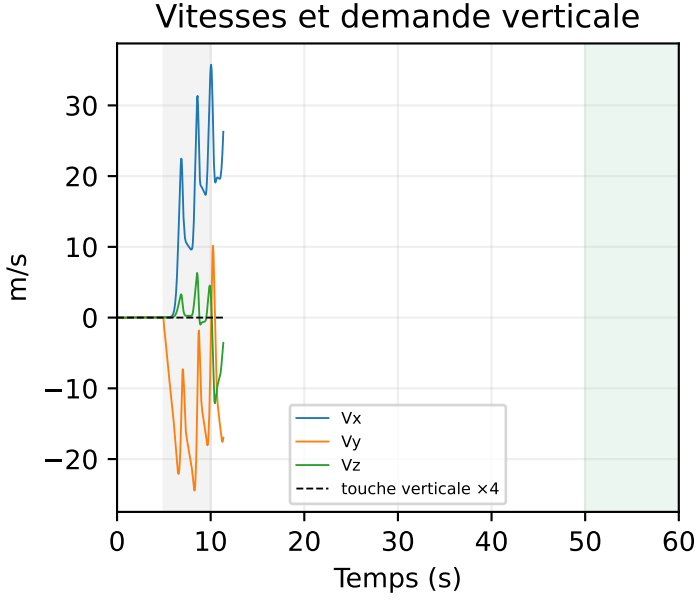
slow-radio / limited / recovery-inverted — FAIL

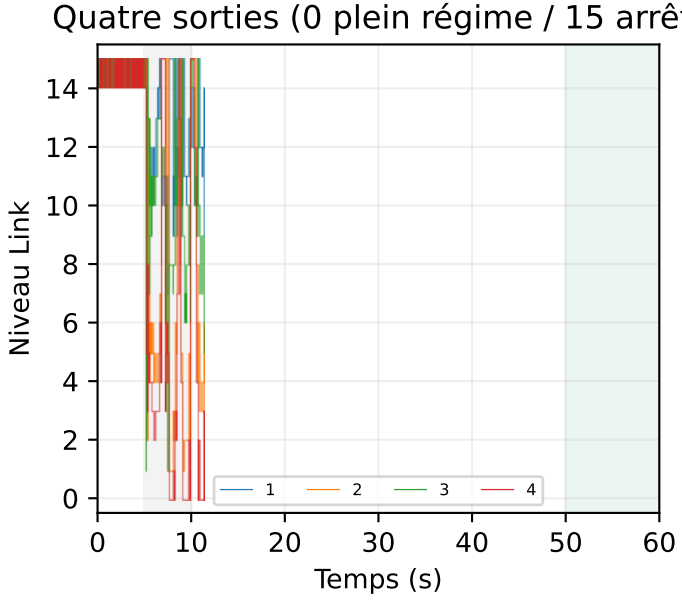
Aucune touche ; inversion instantanée de 180° à 5 s. Perte d’altitude publiée.

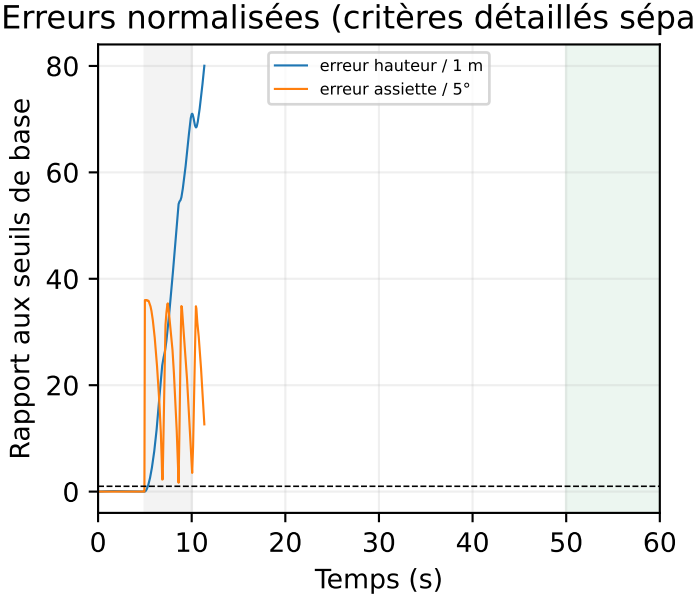












Erreur finale hauteur 80.039 m ; excursion finale — m ; assiette finale max —° ; freinage — s. Critères échoués : complete, tail_height_peak, tail_height_rms, tail_height_span, tail_tilt, tail_speed_rms, final_speed, transient_height, recovery. Arrêt : Declared simulation envelope crossed (ground y=0 / atmosphere y=315 / horizontal 5000m); no contact modeled

Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.